

ASSIGNMENT 01

1. Display current working directory.
2. List all files and hidden files in current directory.
3. Create a directory named documents .
4. Create subdirectories docs , backup , and scripts .
5. Create files:
file1.txt
file2.txt
notes.txt
6. Copy file1.txt into backup .
7. Move notes.txt into docs .
8. Rename file2.txt to data.txt .
9. Delete data.txt .
10. Remove backup directory recursively.

```
ac_163_184@dac:~/Batch2/Assignment01$ pwd
/home/ac_163_184/Batch2/Assignment01
ac_163_184@dac:~/Batch2/Assignment01$ ls -la
total 24
drwxrwxr-x 2 ac_163_184 ac_163_184 4096 May 19 20:30 .
drwxrwxr-x 5 ac_163_184 ac_163_184 4096 May 20 22:45 ..
-rw-rw-r-- 1 ac_163_184 ac_163_184  26 May 19 20:30 error.txt
-rwxr-xr-- 1 ac_163_184 sudo      93 May 19 20:11 file01.txt
-rw-rw-r-- 1 ac_163_184 ac_163_184 130 May 19 20:26 list.txt
-rw-rw-r-- 1 ac_163_184 ac_163_184  72 May 19 20:30 log.txt
ac_163_184@dac:~/Batch2/Assignment01$ mkdir documents
ac_163_184@dac:~/Batch2/Assignment01$ cd documents
ac_163_184@dac:~/Batch2/Assignment01/documents$ mkdir docs backup scripts
ac_163_184@dac:~/Batch2/Assignment01/documents$ touch file1.txt file2.txt notes.txt
ac_163_184@dac:~/Batch2/Assignment01/documents$ cp file1.txt backup/
ac_163_184@dac:~/Batch2/Assignment01/documents$ mv notes.txt docs/
ac_163_184@dac:~/Batch2/Assignment01/documents$ mv file2.txt data.txt
ac_163_184@dac:~/Batch2/Assignment01/documents$ rm data.txt
ac_163_184@dac:~/Batch2/Assignment01/documents$ rm -r backup
```

11. Display contents of example.txt .
12. Display first 10 lines of /etc/passwd .
13. Display last 10 lines of /etc/passwd .
14. Count lines, words, and characters in file.
15. Search word Linux inside file.
16. Search recursively for word error inside /var/log .
17. Find all .txt files in home directory.
18. Find empty files inside /tmp .

```

ac_163_184@dac:~/Batch2/Assignment01/documents$ cat example.txt
this
is
a
sample
text
file
ac_163_184@dac:~/Batch2/Assignment01/documents$ cd /
ac_163_184@dac:/$ cd etc
ac_163_184@dac:/etc$ head -n 10 passwd.txt
head: cannot open 'passwd.txt' for reading: No such file or directory
ac_163_184@dac:/etc$ head -n 10 passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
ac_163_184@dac:/etc$ tail -n 10 passwd
gnome-initial-setup:x:125:65534::/run/gnome-initial-setup:/bin/false
hplip:x:126:7:HPLIP system user,,,:/run/hplip:/bin/false
gdm:x:127:133:Gnome Display Manager:/var/lib/gdm3:/bin/false
cdac:x:1001:1001::/home/cdac:/bin/sh
mysql:x:128:137:MySQL Server,,,:/nonexistent:/bin/false
mongodb:x:129:65534::/home/mongodb:/usr/sbin/nologin
pgcp-ac:x:1000:1000::,/home/pgcp-ac:/bin/bash
fwupd-refresh:x:130:139:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
ac_163_184:x:1002:1003::,/home/ac_163_184:/bin/bash
b1_090:x:1003:1004:krishna,,,:/home/b1_090:/bin/bash
ac_163_184@dac:/etc$ wc -l -w -m passwd
  53   93 3139 passwd
ac_163_184@dac:/etc$ cd Batch2/Assignment01/documents
bash: cd: Batch2/Assignment01/documents: No such file or directory
ac_163_184@dac:/etc$ cd ~
ac_163_184@dac:~$ cd Batch2/Assignment01/documents
ac_163_184@dac:~/Batch2/Assignment01/documents$ nano example.txt
ac_163_184@dac:~/Batch2/Assignment01/documents$ grep "linux" example.txt
linux
ac_163_184@dac:~/Batch2/Assignment01/documents$ nano example.txt
ac_163_184@dac:~/Batch2/Assignment01/documents$ grep -r "error" example.txt
error
ac_163_184@dac:~/Batch2/Assignment01/documents$ ls -a *.txt
example.txt  file1.txt
ac_163_184@dac:~/Batch2/Assignment01/documents$ cd ..
ac_163_184@dac:~/Batch2/Assignment01$ mkdir tmp
ac_163_184@dac:~/Batch2/Assignment01$ cd tmp
ac_163_184@dac:~/Batch2/Assignment01/tmp$ touch 1.txt 2.txt
ac_163_184@dac:~/Batch2/Assignment01/tmp$ find /tmp -type f -empty

```

19. Find files modified within last 2 days.
20. Display size of report.pdf in KB.
21. Display file permissions using long listing format.
22. Give execute permission to script.sh .
23. Remove write permission from others for data.txt .
24. Set permission 754 on project.sh .
25. Change owner of file to another user.
26. Change group ownership of file.

27. Create new user john .
28. Set password for john .
29. Delete user alice with home directory.

```
ac_163_184@dac:~/Batch2/Assignment01$ find . -type f -mtime -2
./file01.txt
./list.txt
./error.txt
./documents/example.txt
./documents/docs/notes.txt
./documents/file1.txt
./log.txt
./tmp/1.txt
./tmp/2.txt
ac_163_184@dac:~/Batch2/Assignment01$ ls -s --block-size=1K report.pdf
324 report.pdf
ac_163_184@dac:~/Batch2/Assignment01$ ls -l
total 348
drwxrwxr-x 4 ac_163_184 ac_163_184 4096 May 21 00:20 documents
-rw-rw-r-- 1 ac_163_184 ac_163_184 26 May 19 20:30 error.txt
-rwxr-xr-- 1 ac_163_184 sudo 93 May 19 20:11 file01.txt
-rw-rw-r-- 1 ac_163_184 ac_163_184 130 May 19 20:26 list.txt
-rw-rw-r-- 1 ac_163_184 ac_163_184 72 May 19 20:30 log.txt
-rw-rw-r-- 1 ac_163_184 ac_163_184 327752 Mar 1 01:51 report.pdf
drwxr-xr-- 2 ac_163_184 ac_163_184 4096 May 21 00:26 tmp
ac_163_184@dac:~/Batch2/Assignment01$ chmod +x script.sh
chmod: cannot access 'script.sh': No such file or directory
ac_163_184@dac:~/Batch2/Assignment01$ touch script.sh
ac_163_184@dac:~/Batch2/Assignment01$ chmod +x script.sh
ac_163_184@dac:~/Batch2/Assignment01$ touch data.txt
ac_163_184@dac:~/Batch2/Assignment01$ chmod o-w data.txt
ac_163_184@dac:~/Batch2/Assignment01$ touch project.sh
ac_163_184@dac:~/Batch2/Assignment01$ chmod 754 project.sh
ac_163_184@dac:~/Batch2/Assignment01$ sudo chown username project.sh
[sudo] password for ac_163_184:
chown: invalid user: 'username'
ac_163_184@dac:~/Batch2/Assignment01$ sudo chown pgcp-ac project.sh
ac_163_184@dac:~/Batch2/Assignment01$ sudo useradd john
ac_163_184@dac:~/Batch2/Assignment01$ sudo passwd john
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
ac_163_184@dac:~/Batch2/Assignment01$ sudo userdel -r john
userdel: john mail spool (/var/mail/john) not found
userdel: john home directory (/home/john) not found
ac_163_184@dac:~/Batch2/Assignment01$ sudo adduser alice
```

30. Display currently logged in users.

```
ac_163_184@dac:~/Batch2/Assignment01$ who
ac_163_184 tty2 2026-05-20 22:28 (tty2)
```

31. Display all environment variables.

```
ac_163_184@dac: ~/Batch2/Assignment01
ac_163_184@dac:~/Batch2/Assignment01$ env
SHELL=/bin/bash
SESSION_MANAGER=local/dac:@/tmp/.ICE-unix/5392,unix/dac:/tmp/.ICE-unix/5392
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
SSH_AGENT_LAUNCHER=gnome-keyring
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
LANGUAGE=en_IN:en
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1002/keyring/ssh
XMODIFIERS=@im=ibus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
PWD=/home/ac_163_184/Batch2/Assignment01
LOGNAME=ac_163_184
XDG_SESSION_DESKTOP=ubuntu
XDG_SESSION_TYPE=wayland
SYSTEMD_EXEC_PID=5421
XAUTORITY=/run/user/1002/.mutter-Xwaylandauth.7S4LP3
HOME=/home/ac_163_184
USERNAME=ac_163_184
IM_CONFIG_PHASE=1
LANG=en_IN
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01
:ni=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=0
1;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31
:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;
31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.d
eb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01
;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.
esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.p
gm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=0
1;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
5:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.
nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=
01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.
ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3
=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;
36:
XDG_CURRENT_DESKTOP=ubuntu:GNOME
VTE_VERSION=6800
WAYLAND_DISPLAY=wayland-0
GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/f70a8bbe_8578_4dae_a421_1e15c7f07839
GNOME_SETUP_DISPLAY=:1
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=ac_163_184
GNOME_TERMINAL_SERVICE=:1.137
DISPLAY=:0
SHLVL=1
QT_IM_MODULE=ibus
XDG_RUNTIME_DIR=/run/user/1002
```

32. Create custom shell variable and print it.
33. Change PS1 prompt temporarily.
34. Redirect output of ls -l into list.txt .
35. Append current date into log.txt .
36. Redirect standard error into error.txt .
37. Redirect both standard output and standard error into same file.

38. Use pipe to count number of running processes.
39. Display only .txt files using grep .
40. Sort contents of a text file.
41. Remove duplicate lines from file.
42. Compare two files using diff .
43. Merge two text files into one file.
44. Reverse contents of file.
45. Compress directory using tar .
46. Extract .tar.gz archive.
47. Display line numbers while viewing file.
48. Append text into file using echo .
49. Display contents of multiple files together.
50. Display current date and time in custom format.

```

ac_163_184@dac:~$ MY_VAR="Hello"
echo $MY_VAR
Hello
ac_163_184@dac:~$ ls -l > list.txt
ac_163_184@dac:~$ date >> log.txt
ac_163_184@dac:~$ command_here 2> error.txt
ac_163_184@dac:~$ command_here > output.txt 2>&1
ac_163_184@dac:~$ ps aux | wc -l
355
ac_163_184@dac:~$ ls | grep "\.txt$"
error.txt
list.txt
log.txt
output.txt
ac_163_184@dac:~$ sort filename.txt
sort: cannot read: filename.txt: No such file or directory
ac_163_184@dac:~$ sort example.txt
sort: cannot read: example.txt: No such file or directory
ac_163_184@dac:~$ sort filename.txt | uniq
sort: cannot read: filename.txt: No such file or directory
ac_163_184@dac:~$ diff log.txt error.txt
1c1
< Thursday 21 May 2026 12:39:44 AM IST
---
> command_here: command not found
ac_163_184@dac:~$ cat log.txt error.txt > merged.txt
ac_163_184@dac:~$ tac merged.txt
command_here: command not found
Thursday 21 May 2026 12:39:44 AM IST
ac_163_184@dac:~$ tar -czf archive.tar.gz directory_name/
tar: directory_name: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
ac_163_184@dac:~$ echo "Text to add" >> filename.txt
ac_163_184@dac:~$ cat list.txt log.txt error.txt
total 44
drwxrwxr-x 5 ac_163_184 ac_163_184 4096 May 20 22:45 Batch2
drwxr-x-- 5 ac_163_184 ac_163_184 4096 May 12 04:44 chrome_local
drwxr-xr-x 2 ac_163_184 ac_163_184 4096 May 19 19:46 Desktop
drwxr-xr-x 2 ac_163_184 ac_163_184 4096 May 19 19:46 Documents
drwxr-xr-x 3 ac_163_184 ac_163_184 4096 May 19 19:47 Downloads
-rw-rw-r-- 1 ac_163_184 ac_163_184 0 May 21 00:39 list.txt
drwxr-xr-x 2 ac_163_184 ac_163_184 4096 May 19 19:46 Music
drwxr-xr-x 3 ac_163_184 ac_163_184 4096 May 19 20:17 Pictures
drwxr-xr-x 2 ac_163_184 ac_163_184 4096 May 19 19:46 Public
drwx----- 4 ac_163_184 ac_163_184 4096 May 19 19:46 snap
drwxr-xr-x 2 ac_163_184 ac_163_184 4096 May 19 19:46 Templates
drwxr-xr-x 2 ac_163_184 ac_163_184 4096 May 19 19:46 Videos
Thursday 21 May 2026 12:39:44 AM IST
command_here: command not found
ac_163_184@dac:~$ date "+%Y-%m-%d %H:%M:%S"
2026-05-21 00:42:04
ac_163_184@dac:~$ █

```

